INCH-POUND

MIL-M-38510/50E
AMENDMENT 2
16 August 1988
SUPERSEDING
AMENDMENT 1
31 January 1985

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, CMOS, NAND GATES
MONOLITHIC SILICON, POSITIVE LOGIC

This amendment forms a part of MIL-M-38510/50E, dated 30 April 1984, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* 1.2.3, add new case outline as follows:

"T F-2A (14-lead, .390" x .260" x .115"), flat package".

PAGE 3

4.2a: Delete "3.1.9 through 3.1.13" and substitute "Initial (pre burn-in) electrical parameters (3.1.10) through interim (post burn-in) electrical parameters (3.1.14)".

PAGE 4

* 4.4.1, add the following new subparagraphs:

"d. Subgroups 9 and 11 shall be measured only for initial qualification and after process or design changes which may affect dynamic performance.

"e. When device types 01 through 03 are qualified by extension (see 4.3.1), these device types will be inspected (QCI) according to the requirements for device types 51 through 53, respectively."

PAGE 5

TABLE I, V_{OL2}, Min column: Delete "0.05". Max column: Add "0.05".

TABLE I, V_{OL3}, Min column: Delete "1.25". Max column: Add "1.25".

PAGE 7

* TABLE I, t_{PHL} and t_{PLH}, Min limits column: Delete all entries and substitute "13".

* TABLE I, t_{THL} and t_{TLH}, Min limits column: Delete all entries and substitute "10".

PAGE 8

* TABLE II, Line no. 9, Class B devices, Table III subgroups, add: "10,11".

* TABLE II, Line no. 12: Delete in its entirety.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:03Z
Check the source to verify that this is the current version before use.

PAGE 9

* FIGURE 1: Delete "Cases A, C, D, X and Y" (three places) and substitute "Cases A, C, D, T, X, and Y" (three places).

PAGE 11

FIGURE 3, note 3: Delete "2 kΩ or 47 kΩ" and substitute "2 kΩ to 47 kΩ".

PAGE 12

FIGURE 4, note 1: Delete "2 kΩ or 47 kΩ" and substitute "2 kΩ to 47 kΩ".

PAGE 14

* TABLE III: Delete "Cases A, C, D, X, Y" and substitute "Cases A, C, D, T, X, Y".

PAGE 15

* TABLE III: Delete "Cases A, C, D, X, Y" and substitute "Cases A, C, D, T, X, Y".

* TABLE III, t_{PHL} and t_{PLH}, Min limits column: Delete "20", "30", "15" and substitute "13", "18", "13", respectively.

PAGE 16

* TABLE III: Delete "Cases A, C, D, X, Y" and substitute "Cases A, C, D, T, X, Y".

* TABLE III, t_{PHL} and t_{PLH}, Min limits column: Delete "20", "30", "15" and substitute "13", "18", "13", respectively.

* TABLE III, t_{THL}, Min limits column: Delete "30", "45", "25" and substitute "10", "14", "10", respectively. t_{TLH}, Min limits column: Delete "40", "60", "30" and substitute "10", "14", "10", respectively.

PAGE 17

* TABLE III: Delete "Cases A, C, D, X, Y" and substitute "Cases A, C, D, T, X, Y".

PAGE 18

* TABLE III: Delete "Cases A, C, D, X, Y" and substitute "Cases A, C, D, T, X, Y".

* TABLE III, t_{PHL} and t_{PLH}, Min limits column: Delete "20", "30", "15" and substitute "13", "18", "13", respectively.

PAGE 19

* TABLE III: Delete "Cases A, C, D, X, Y" and substitute "Cases A, C, D, T, X, Y".

* TABLE III, t_{PHL} and t_{PLH}, Min limits column: Delete "20", "30", "15" and substitute "13", "18", "13", respectively.

* TABLE III, t_{THL}, Min limits column: Delete "50", "70", "40" and substitute "10", "14", "10", respectively. t_{TLH}, Min limits column: Delete "40", "60", "35" and substitute "10", "14", "10", respectively.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:03Z
Check the source to verify that this is the current version before use.

PAGES 20 and 21

* TABLE III: Delete "Cases A, C, D, X, Y" and substitute "Cases A, C, D, T, X, Y".

PAGE 22

* TABLE III: Delete "Cases A, C, D, X, Y" and substitute "Cases A, C, D, T, X, Y".

* TABLE III, t_{PHL} and t_{PLH}, Min limits column: Delete "20", "30", "15" and substitute "13", "18", "13", respectively. t_{THL}, Min limits column: Delete "30", "45", "25" and substitute "10", "14", "10", respectively. t_{TLH}, Min limits column: Delete "40", "60", "30" and substitute "10", "14", "10", respectively.

PAGES 23 through 25

* TABLE III: Delete "Cases A, C, D, X, Y" and substitute "Cases A, C, D, T, X, Y".

PAGE 26

* TABLE III: Delete "Cases A, C, D, X, Y" and substitute "Cases A, C, D, T, X, Y".

* TABLE III, t_{PHL}, Min limits column: Delete "10", "15", "10" and substitute "13", "18", "13", respectively. t_{PLH}, Min limits column: Delete "8", "11", "8" and substitute "13", "18", "13", respectively. t_{THL} and t_{TLH}, Min limits column: Delete "15", "23", "15" and substitute "10", "14", "10", respectively.

PAGES 27 and 28

* TABLE III: Delete "Cases A, C, D, X, Y" and substitute "Cases A, C, D, T, X, Y".

PAGE 29

* TABLE III: Delete "Cases A, C, D, X, Y" and substitute "Cases A, C, D, T, X, Y".

* TABLE III, t_{PHL}, Min limits column: Delete "16", "25", "15" and substitute "13", "18", "13", respectively. t_{PLH1}, Min limits column: Delete "19" and substitute "18".

PAGE 30

* TABLE III: Delete "Cases A, C, D, X, Y" and substitute "Cases A, C, D, T, X, Y".

* TABLE III, t_{THL}, Min limits column: Delete "28", "41", "28" and substitute "10", "14", "10", respectively. t_{TLH}, Min limits column: Delete "21", "32", "21" and substitute "10", "14", "10", respectively.

PAGES 31 and 32

* TABLE III: Delete "Cases A, C, D, X, Y" and substitute "Cases A, C, D, T, X, Y".

Source: https://assist.dla.mil -- Downloaded: 202610-21T13:03Z
Check the source to verify that this is the current version before use.

PAGE 33

* TABLE III: Delete "Cases A, C, D, X, Y" and substitute "Cases A, C, D, T, X, Y".

* TABLE III, t_{PHL}, Min limits column: Delete "9", "13", "9" and substitute "13", "18", "13", respectively.

PAGE 34

* TABLE III: Delete "Cases A, C, D, X, Y" and substitute "Cases A, C, D, T, X, Y".

* TABLE III, t_{PLH}, Min limits column: Delete "8", "11", "8" and substitute "13", "18", "13", respectively. t_{THL}, Min limits column: Delete "19" and substitute "18". t_{TLH}, Min limits column: Delete "15", "25", "12" and substitute "13", "18", "13", respectively.

PAGE 35

* 4.4.3c, delete in its entirety and substitute the following:

"c. When device types 01 through 03 are qualified by extension (see 4.3.1), these device types will be inspected (QCI) according to the requirements for device types 51 through 53, respectively."

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - MI
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - AR, SM
Navy - AS, CG, MC, OC, SH

Preparing activity:
NASA - NA

Agent:
DLA - ES

(Project 5962-1156-1)

Source: https://assist.dla.mil -- Downloaded: 202610-21T13:03Z
Check the source to verify that this is the current version before use.